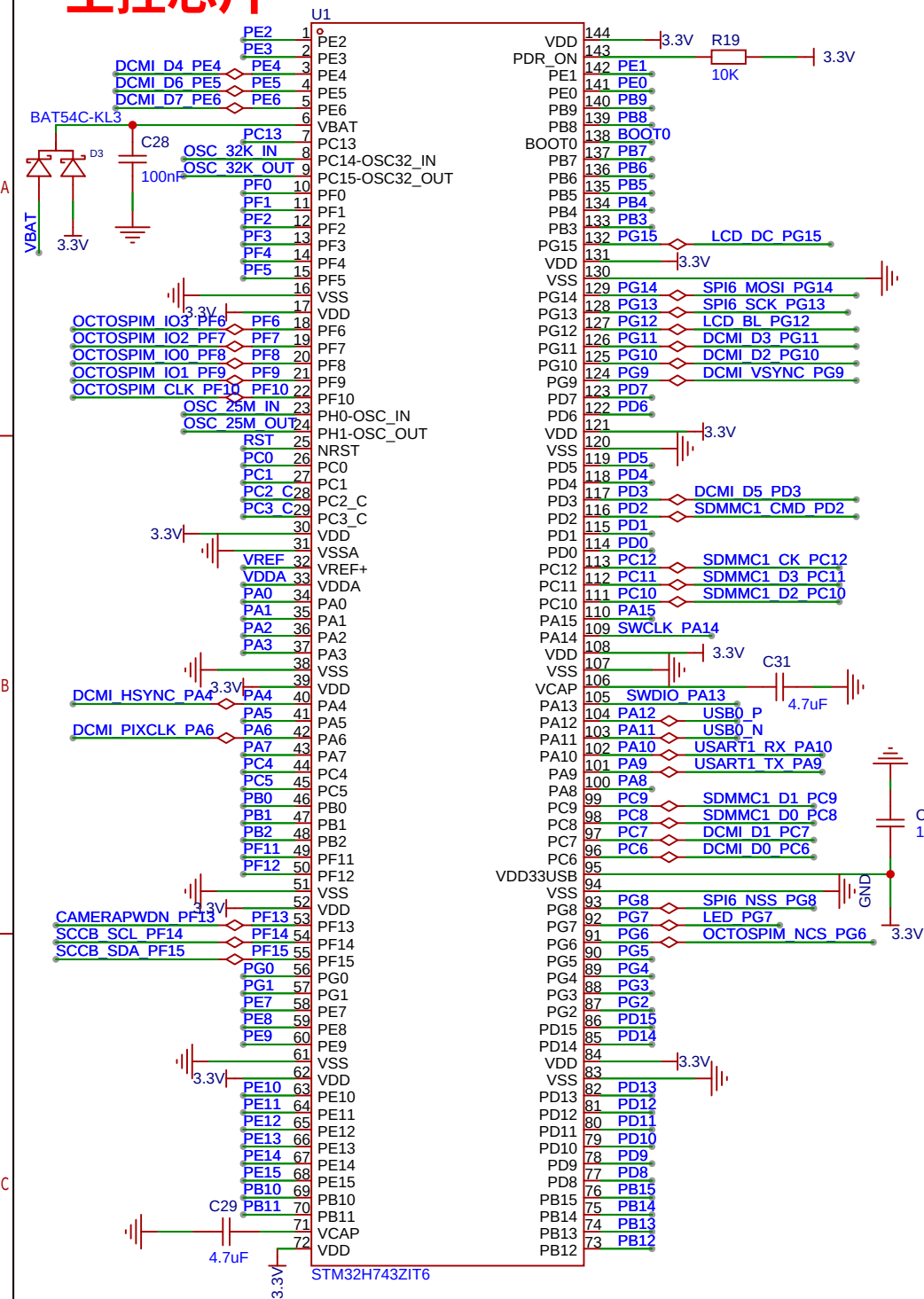
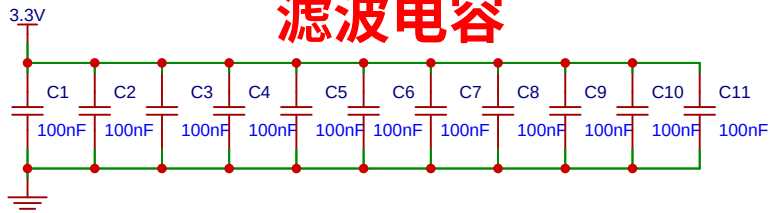


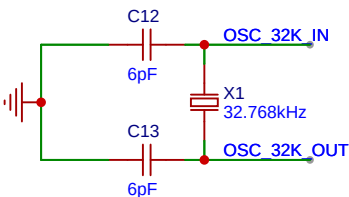
主控芯片



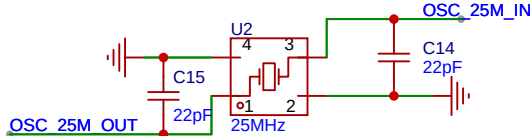
滤波电容



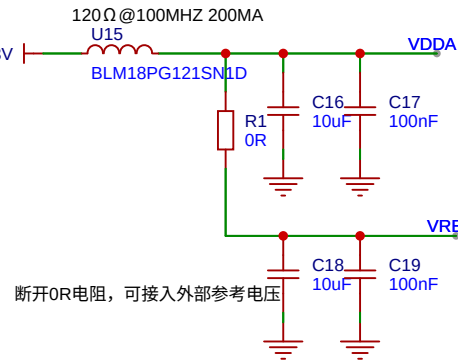
RTC时钟晶振32.768KHz



25M主时钟晶振



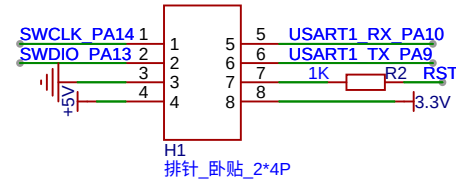
模拟电压



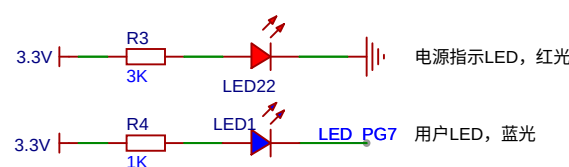
参考电压

断开0R电阻，可接入外部参考电压

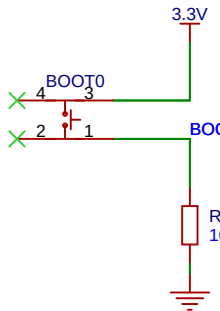
SWD和USART1接口



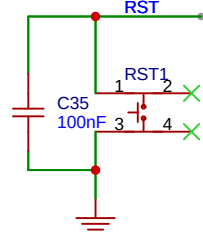
- 1.此处的1K电阻目的是在用户将电源线错接到RST时起到限流保护作用
2. 推荐使用5V供电，再经过板载的稳压芯片得到3.3V给单片机，避免因意外输入高压而损坏单片机



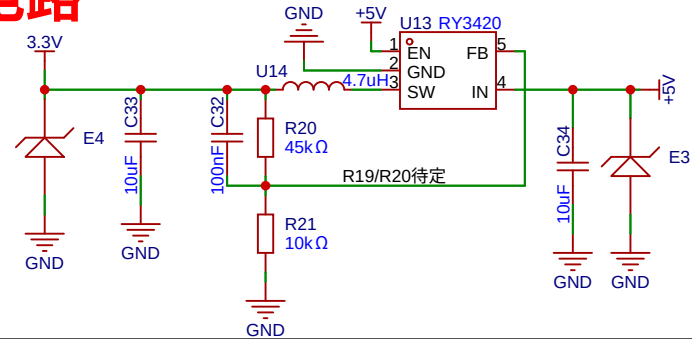
BOOT0电路



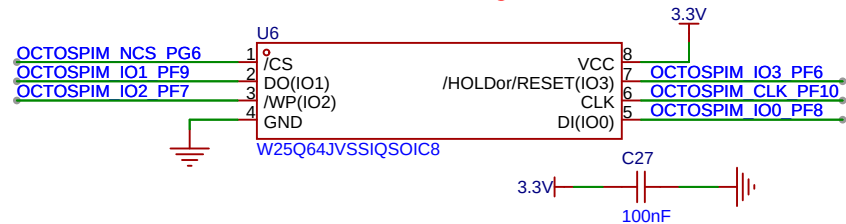
复位电路



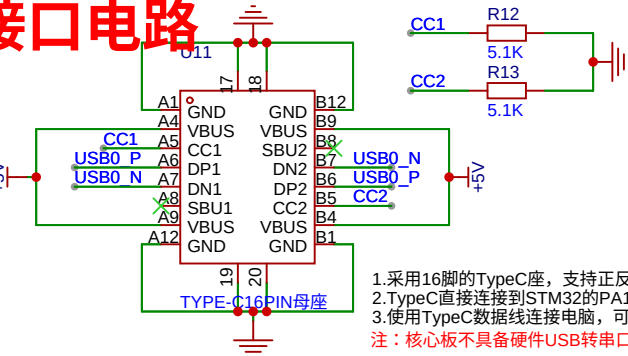
3V3电源电路



扩展Flash W25Q64

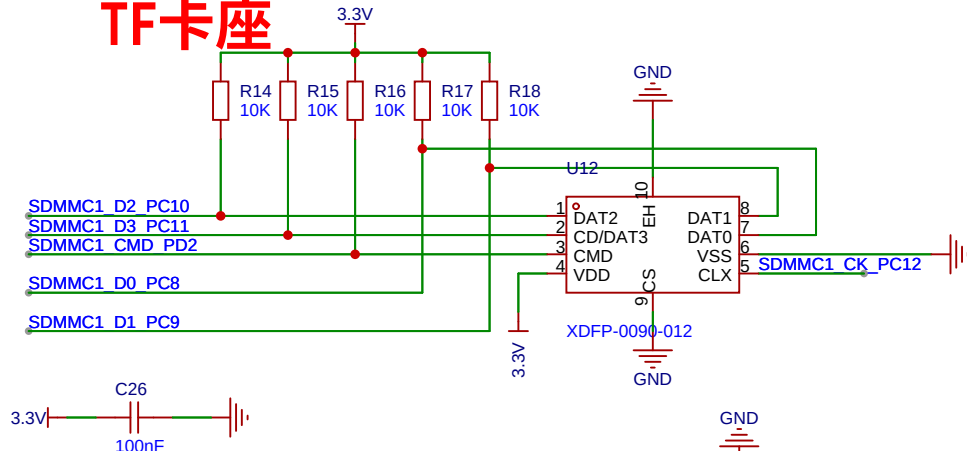


USB接口电路

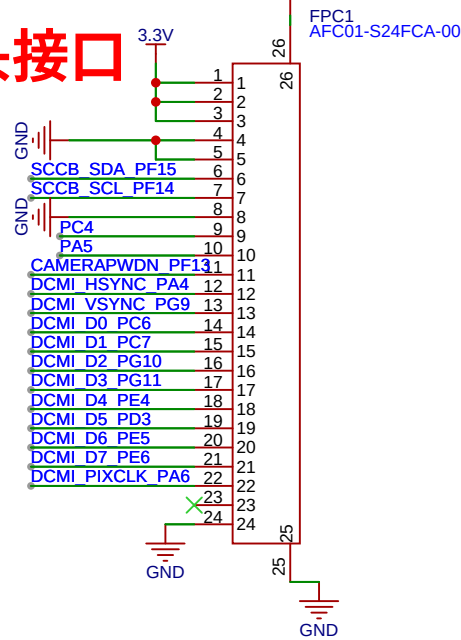


- 1.采用16脚的TypeC座，支持正反插
 - 2.TypeC直接连接到STM32的PA11和PA12，可开发USB应用
 - 3.使用TypeC数据线连接电脑，可进行USB DFU下载用户程序
- 注：核心板不具备硬件USB转串口电路！！

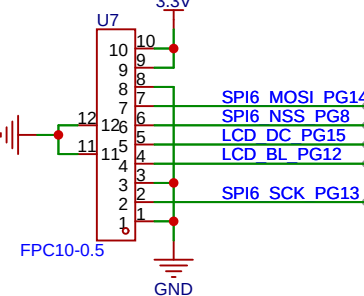
TF卡座



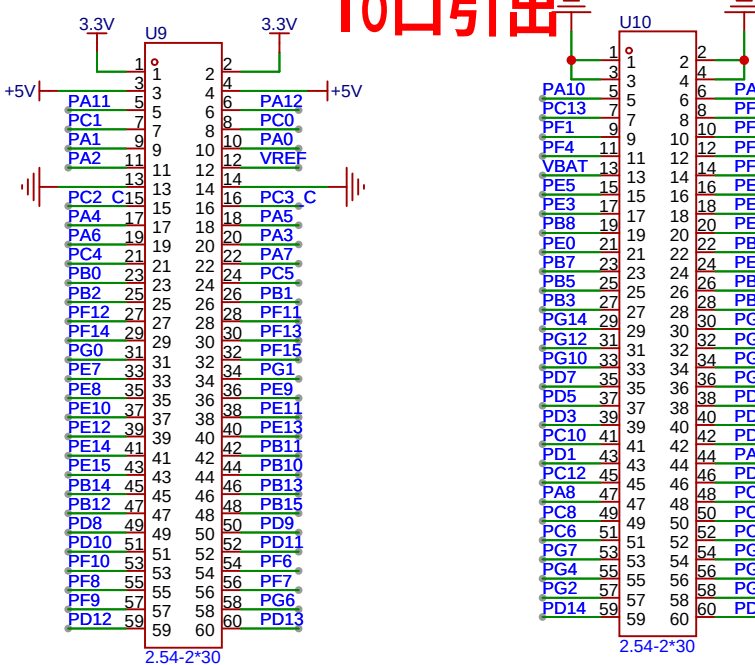
DCMI摄像头接口



SPI液晶接口



I/O口引出



使用核心板驱动电机、高电压或高干扰的设备时，一定要加光耦隔离，否则很容易损坏单片机！！